

Day 4 Firewall theory

1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

app.

Steps to Block Spotify Using Windows Defender Firewall

Press Windows + R, type wf.msc, and press Enter.

In Windows Defender Firewall with Advanced Security, click Outbound Rules.

Click New Rule... on the right side.

Select Program → Next.

Select This program path → Browse and select the Spotify.exe file.

Common location:

C:\Users\<YourName>\AppData\Roaming\Spotify\Spotify.exe

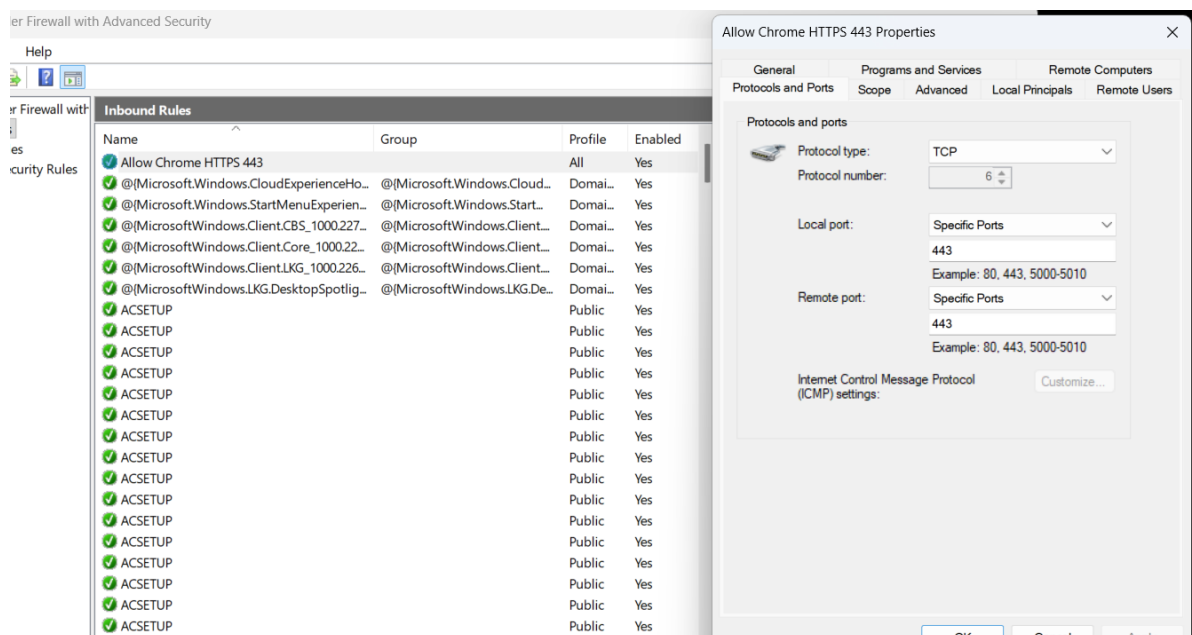
Select Block the connection → Next.

Keep Domain, Private, and Public checked → Next.

Enter a name such as Block Spotify Outbound Traffic → Finish.

The new rule will appear under Outbound Rules and should show Enabled: Yes and Action: Block.

2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections onfor Google Chrome on port 443 (HTTPS).



Inbound Firewall Rule for Google Chrome – Port 443

I set up an inbound rule in **Windows Defender Firewall** to allow Google Chrome to receive incoming **HTTPS connections on TCP port 443**.

Configuration:

- **Rule Type:** Custom
- **Program:** chrome.exe
- **Protocol:** TCP
- **Local Port:** 443
- **Remote Port:** All Ports
- **Action:** Allow the connection
- **Profile:** Domain, Private, Public
- **Rule Name:** Allow Chrome HTTPS 443

Result:

The firewall rule allows incoming TCP connections for **Google Chrome on port 443 (HTTPS)**.

3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

Block All Traffic from a Specific IP Address

Example IP address: **123.45.67.89**

Step-by-step

1. Press **Windows + R**, type wf.msc, and press **Enter**.
2. Select **Inbound Rules** from the left panel.
3. Click **New Rule...**
4. Select **Custom** → **Next**.
5. Under **Program**, select **All programs** → **Next**.
6. Under **Protocol and Ports**, keep:
 - Protocol type: **Any**
 - Local port: **All Ports**
 - Remote port: **All Ports**

Click **Next**.

7. On the **Scope** page, find **Remote IP address**.

8. Select **These IP addresses** → **Add...**

9. Enter:

123.45.67.89

Click **OK** → **Next**.

10. Select **Block the connection** → **Next**.

11. Keep **Domain, Private, and Public** checked → **Next**.

12. Give the rule a name:

Block IP 123.45.67.89

13. Click **Finish**.

Final Configuration

Rule Name: Block IP 123.45.67.89

Program: All programs

Protocol: Any

Remote IP: 123.45.67.89

Local Ports: All

Remote Ports: All

Action: Block

Profiles: Domain, Private, Public

Conclusion:

The firewall rule blocks all incoming traffic from the specified IP address **123.45.67.89** regardless of the port or application.

4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

Difference	Hardware Firewall	Software Firewall
Location	Separate physical device placed between the network and Internet	Installed directly on a computer/server
Protection	Protects multiple devices across the entire network	Mainly protects the individual device

Difference	Hardware Firewall	Software Firewall
Performance	Designed to handle large amounts of network traffic	Uses the computer's CPU and memory resources
Use Case	Large offices, companies, schools, and data centers	Personal computers, laptops, and individual servers

Use Cases

- **Hardware Firewall:** A large office uses a hardware firewall at the Internet gateway to filter traffic before it reaches employees' computers.
- **Software Firewall: Windows Defender Firewall** runs on a Windows laptop to control incoming and outgoing connections for applications.